CSE310: Compiler Sessional, January 2026 Term
ICG Online Assignment Practice
Logical XOR Operator ^^

Time: 40 Mins **Total Marks: 10**

Problem

Extend the grammar to support a logical XOR operator `^^` for expressions. It should return 1 if exactly one of its operands is true (non-zero), and 0 otherwise.

```
int a, b; a = 1; b = 0; println(a ^^ b);
```

Task

- Add `^^` to the `logic_expression` grammar rule.
- Evaluate both operands. Convert them to strict boolean values (0 or 1), then perform a bitwise XOR on the results to yield the final logical XOR value in `EAX`.

Mark Distribution

- Extending the grammar to support `^^`. (4)
- Code generation for logical evaluation and XOR. (6)

Sample Input-Output

Input Program	Generated Sample Assembly (print library omitted)

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```
int main() {
    int a, b;
    a = 1;
    b = 0;
    println(a ^^ b);
    println(a ^^ a);
    return 0;
}
```

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```
main:
    PUSH EBP
    MOV EBP, ESP
    SUB ESP, 8
    MOV EAX, 1
    MOV [EBP-4], EAX
    MOV EAX, 0
    MOV [EBP-8], EAX
    MOV EAX, [EBP-4]
    CMP EAX, 0
    SETNE AL
    MOVZX EAX, AL
    PUSH EAX
    MOV EAX, [EBP-8]
    CMP EAX, 0
    SETNE AL
    MOVZX EDX, AL
    POP EAX
    XOR EAX, EDX
    CALL print_number
    MOV EAX, [EBP-4]
    CMP EAX, 0
    SETNE AL
    MOVZX EAX, AL
    PUSH EAX
    MOV EAX, [EBP-4]
    CMP EAX, 0
    SETNE AL
    MOVZX EDX, AL
    POP EAX
    XOR EAX, EDX
    CALL print_number
    MOV EAX, 0
    JMP .L1
.L1:
    ADD ESP, 8
    POP EBP
    RET
```

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<td>Expected Output</td>
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```
1
0
```

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